

Project Elective: Weekly Progress Report (Config-2)

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1 Introduction

This report details the work completed for the Project Elective during the past week, focusing on **Config-2**. In this configuration, the Transmitter (Tx) Ground electrode is positioned directly above the Tx-Signal electrode, representing a vertical stacking arrangement, which differs from the side-by-side placement in Config-1. This setup aims to investigate the impact of electrode orientation on signal transmission through human tissue.

2 FEMM Simulation Code for Config-2

The following Python script was developed for Config-2. Key modifications include the coordinate definitions for the Tx-Signal and Tx-Ground electrodes to reflect their new vertical arrangement.

```
1 # -*- coding: utf-8 -*-
2 """
3 Config-2: Tx-Ground electrode positioned ABOVE Tx-Signal electrode
4 """
5
6 import femm
7 import csv
8
9 # === Helper functions ===
10 def assign_segments(boundary=None, conductor=None, points=[]):
11     for x, y in points:
12         femm.ei_selectsegment(x, y)
13         femm.ei_setsegmentprop(boundary or '<None>', 1, 1, 0, 0,
14                               conductor or '<None>')
15         femm.ei_clearselected()
16
17 def label_block(x, y, material):
18     femm.ei_addblocklabel(x, y)
19     femm.ei_selectlabel(x, y)
20     femm.ei_setblockprop(material, 1, 1, 0)
21     femm.ei_clearselected()
22
23 def get_rectangle_edges(x1, y1, x2, y2):
24     return [
25         ((x1 + x2) / 2, y1),
```

```

25         ((x1 + x2) / 2, y2),
26         (x1, (y1 + y2) / 2),
27         (x2, (y1 + y2) / 2),
28     ]
29
30 def get_rectangle_center(x1, y1, x2, y2):
31     return (x1 + x2) / 2, (y1 + y2) / 2
32
33 def get_rectangle_topleft(x1, y1, x2, y2):
34     left, right = min(x1, x2), max(x1, x2)
35     bottom, top = min(y1, y2), max(y1, y2)
36     dx = 0.01 * (right - left)
37     dy = 0.01 * (top - bottom)
38     return (left + dx, top - dy)
39
40 # === Main FEMM model function ===
41 def run_with_params(params):
42     femm.openfemm()
43     femm.newdocument(1)
44     femm.ei_probdef('centimeters', 'planar', 1e-8, 7, 30)
45
46     TxD = params['TxD']
47     TxW = params['TxW']
48     TxG = params['TxG']
49     TxT = params['TxT']
50     RxT = params['RxT']
51     RxH = params['RxH']
52     RxW = params['RxW']
53
54     # === CRITICAL CHANGE FOR CONFIG-2 ===
55     # In Config-1: Tx-Ground is to the LEFT of Tx-Signal (x-direction)
56     # In Config-2: Tx-Ground is ABOVE Tx-Signal (y-direction)
57
58     rectangles = {
59         "Head": (-12.5, 19.4, -1.5, 8.5),
60         "Body": (-14, 8.4, 0, -38.9),
61         "Leg": (-12, -39.0, -2, -109),
62
63         "Skin-Bottom": (0, 0.000, 70, 0.063),
64         "Fat-Lower": (0, 0.063, 70, 0.520),
65         "Muscle-Lower": (0, 0.520, 70, 1.775),
66         "Bone-Lower": (0, 1.775, 70, 2.127),
67         "Bone-Marrow": (0, 2.127, 70, 2.500),
68         "Bone-Upper": (0, 2.500, 70, 2.852),
69         "Muscle-Upper": (0, 2.852, 70, 4.107),
70         "Fat-Upper": (0, 4.107, 70, 4.564),
71         "Skin-Top": (0, 4.564, 70, 4.627),
72
73         # CONFIG-2: Tx-Signal is BELOW Tx-Ground (vertically stacked)
74         # Tx-Signal sits on top of skin at y = 4.627
75         "Tx-Signal": (70-TxD-TxW, 4.627, 70-TxD, 4.627+TxT),
76
77         # Tx-Ground is ABOVE Tx-Signal, separated by gap TxG
78         "Tx-Ground": (70-TxD-TxW, 4.627+TxT+TxG, 70-TxD, 4.627+TxT+TxG+
79 TxT),
80
81         # Rx electrodes remain the same
82         "Rx-Signal": (55, 0, 70, 0-RxT),

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82     "Rx-Gnd": (55, 0-RxH-2*RxT, 55+RxW, 0-RxH-RxT),
83
84     "Hub": (55, 0-RxT, 55+RxW, 0-RxT-RxH),
85
86     "Wood": (55, 0-RxH-2*RxT, 90, -109),
87     "Outer": (-610, 610, 610, -610)
88 }
89
90 for coords in rectangles.values():
91     femm.ei_drawrectangle(*coords)
92
93 femm.ei_addboundprop('Ground', 0, 0, 0, 0, 0)
94 assign_segments(boundary='Ground', points=get_rectangle_edges(*
rectangles["Outer"]))
95
96 femm.ei_addmaterial('Air', 1, 1, 0)
97 femm.ei_addmaterial('Skin', 2.0e4, 1, 0)
98 femm.ei_addmaterial('Fat', 1.0e2, 1, 0)
99 femm.ei_addmaterial('Muscle', 8.0e3, 1, 0)
100 femm.ei_addmaterial('Bone', 2.5e2, 1, 0)
101 femm.ei_addmaterial('Bone-Marrow', 2.5e2, 1, 0)
102 femm.ei_addmaterial('Wood', 2.0, 1, 0)
103
104 conductor_regions = ["Head", "Body", "Leg", "Tx-Signal", "Tx-Ground",
"Rx-Signal", "Rx-Gnd"]
105
106 material_map = {
107     "Skin-Bottom": "Skin",
108     "Skin-Top": "Skin",
109     "Fat-Lower": "Fat",
110     "Fat-Upper": "Fat",
111     "Muscle-Lower": "Muscle",
112     "Muscle-Upper": "Muscle",
113     "Bone-Upper": "Bone",
114     "Bone-Lower": "Bone",
115     "Bone-Marrow": "Bone-Marrow",
116     "Wood": "Wood",
117     "Hub": "Wood"
118 }
119
120 for name in conductor_regions:
121     if name in rectangles:
122         label_block(*get_rectangle_center(*rectangles[name]), '<No
Mesh>')
123
124 for name, mat in material_map.items():
125     if name in rectangles:
126         label_block(*get_rectangle_center(*rectangles[name]), mat)
127
128 label_block(*get_rectangle_topleft(*rectangles["Outer"]), 'Air')
129
130 femm.ei_addconductorprop('eTx-Signal', 5, 0, 1)
131 femm.ei_addconductorprop('eTx-Ground', 0, 0, 1)
132 femm.ei_addconductorprop('eRx-Signal', 0, 0, 0)
133 femm.ei_addconductorprop('eRx-Gnd', 0, 0, 0)
134 femm.ei_addconductorprop('eHead', 0, 0, 0)
135 femm.ei_addconductorprop('eBody', 0, 0, 0)
136 femm.ei_addconductorprop('eLeg', 0, 0, 0)

```

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137
138     assign_segments(conductor='eTx-Ground', points=get_rectangle_edges
139 (*rectangles["Tx-Ground"]))
140     assign_segments(conductor='eTx-Signal', points=get_rectangle_edges
141 (*rectangles["Tx-Signal"]))
142     assign_segments(conductor='eRx-Gnd', points=get_rectangle_edges(*
143 rectangles["Rx-Gnd"]))
144     assign_segments(conductor='eRx-Signal', points=get_rectangle_edges
145 (*rectangles["Rx-Signal"]))
146     assign_segments(conductor='eHead', points=get_rectangle_edges(*
147 rectangles["Head"]))
148     assign_segments(conductor='eBody', points=get_rectangle_edges(*
149 rectangles["Body"]))
150     assign_segments(conductor='eLeg', points=get_rectangle_edges(*
151 rectangles["Leg"]))
152
153     femm.ei_saveas("temp_sim_config2.fee")
154     femm.ei_analyze(1)
155     femm.ei_loadsolution()
156
157     v_rx_signal = femm.eo_getconductorproperties('eRx-Signal')[0]
158     v_rx_gnd = femm.eo_getconductorproperties('eRx-Gnd')[0]
159     return v_rx_signal - v_rx_gnd
160
161 # === Sweep Controller ===
162 defaults = {
163     'TxD': 20,
164     'TxW': 3,
165     'TxG': 3,
166     'RxH': 30,
167     'TxT': 0.2,
168     'RxT': 0.2,
169     'RxW': 3,
170 }
171
172 sweeps = {
173     'TxD': list(range(10, 41, 31)),
174     'TxW': list(range(1, 11, 11)),
175     'TxG': list(range(1, 11, 11)),
176     'RxH': list(range(10, 51, 41)),
177     'RxW': list(range(5, 35, 30))
178 }
179
180 output_file = "results_config2.csv"
181
182 with open(output_file, mode='w', newline='') as f:
183     writer = csv.writer(f)
184     headers = ["SweepParam"] + list(defaults.keys()) + ["Rx_Voltage"]
185     writer.writerow(headers)
186
187     for param, values in sweeps.items():
188         for val in values:
189             print(f"[INFO] Sweeping {param} = {val}")
190             current_params = defaults.copy()
191             current_params[param] = val
192
193             v_diff_rx = run_with_params(current_params)

```

```

187         writer.writerow([param] + [current_params[key] for key in
188             defaults] + [f"{v_diff_rx:.6f}"])
189 femm.closefemm()

```

Listing 1: Python script for FEMM automation and parameter sweep (Config-2).

3 Simulation Model Image for Config-2

Figure 1 shows a visualization of the FEMM model for Config-2. Note the vertical arrangement of the Tx-Signal and Tx-Ground electrodes as described in the code.

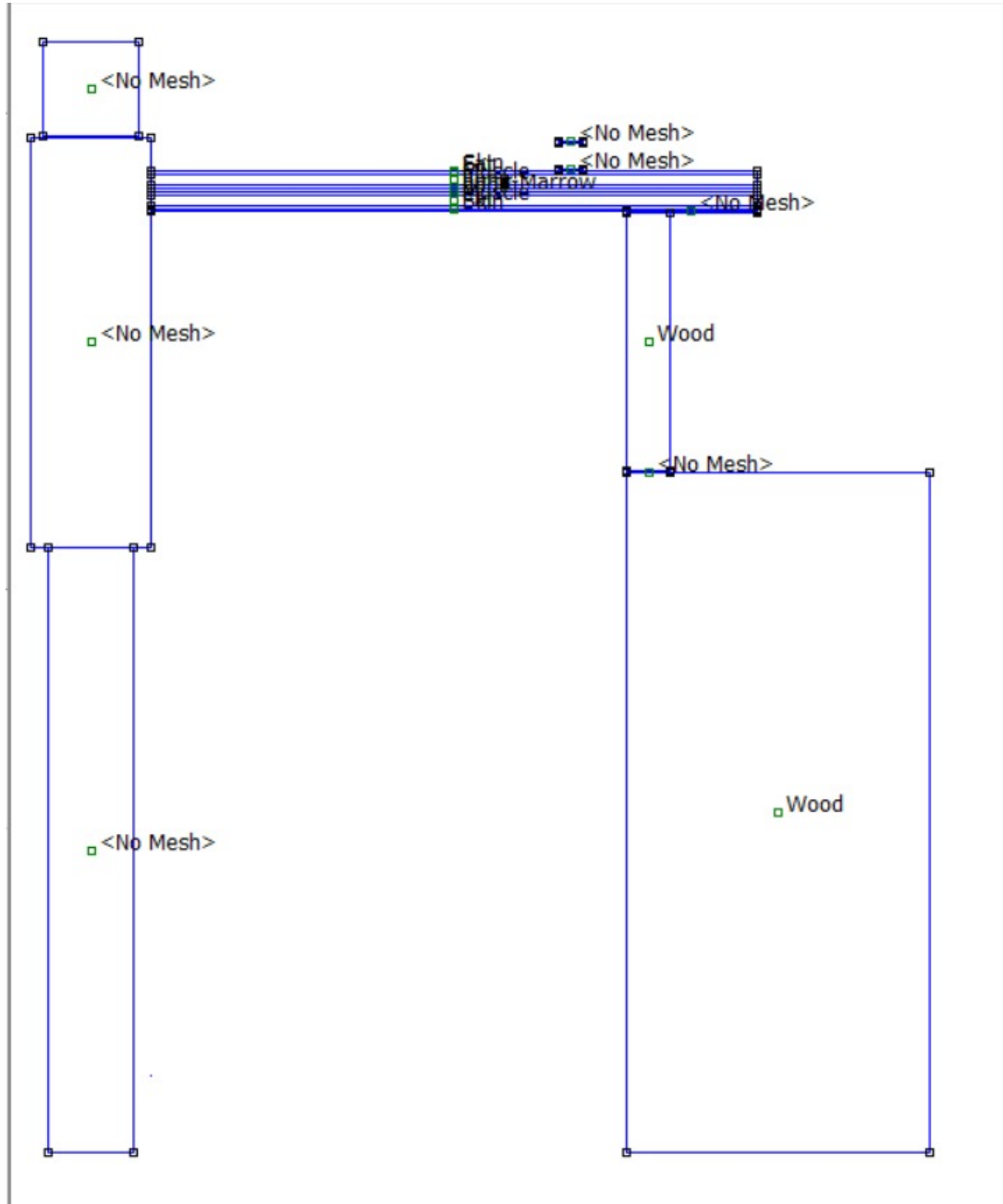


Figure 1: FEMM Simulation Model Visualization (Config-2)